

Leonardo Venturoso

Machine Learning Researcher | Computer Vision and Applied Deep Learning

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PROFESSIONAL SUMMARY

Applied machine learning researcher working on computer vision and deep learning, with a focus on turning state-of-the-art methods into working solutions for real-world problems. Published author and peer reviewer for IEEE Robotics and Automation Letters, with a strong methodological bent: controlled ablations, multi-seed variance analysis, and rigorous benchmarking of CNN and transformer architectures. Two MSc degrees (Data Science; Psychology/Neuroscience) with a deep foundation in statistics and experimental design. Comfortable reading, replicating and adapting recent literature, and sharing it within a collaborative team.

SKILLS

Machine Learning Research: deep learning, computer vision (semantic/instance segmentation, object detection), CNN and transformer architectures (YOLO family, RF-DETR, Mask2Former), graph neural networks, multimodal and polarimetric imaging

Methodology: experimental design, controlled ablations and confound isolation, multi-seed variance analysis, benchmarking under uniform protocols, generalization and domain-shift analysis, reproducibility

Statistics: hypothesis testing, confidence intervals, power and variance analysis, mixed models, high-dimensional time-series analysis

Programming and Tools: Python, PyTorch, Scikit-learn, NumPy, Pandas, R, Git, Docker, Linux

Research Activity: peer review (IEEE Robotics and Automation Letters), academic publications, founder of an internal AI journal club

Languages: Italian (native), English (IELTS C1), German (A1)

PROFESSIONAL EXPERIENCE

Fraunhofer Italia | Bolzano, Italy

Applied AI Researcher | Feb 2024 - Present

- Conduct applied machine-learning research within the Machine Intelligence Development (M.IN.D.) team, transferring state-of-the-art methods to real-world computer-vision problems and evaluating them under rigorous, reproducible protocols.
- Lead a systematic study comparing CNN and transformer segmentation architectures (YOLOv8/v11, RF-DETR-Seg, Mask2Former) across RGB and polarimetric imaging, with controlled ablations isolating input resolution and analysis of generalization under domain shift; results submitted for publication.
- Investigate multimodal and polarimetric imaging, exploiting the physical structure of the light signal as an information channel beyond raw intensity for detection and segmentation.
- Explore graph neural networks for spatio-temporal forecasting on heterogeneous real-world data, from methodology to deployed prototype.
- Peer reviewer for IEEE Robotics and Automation Letters; founded and run the institute's internal AI journal club, keeping the team current with recent deep-learning literature; technical supervision of a small research team on a human-centered AI project.

Fondazione Bruno Kessler (FBK) | Trento, Italy

Data Science Researcher (part-time collaboration) | Mar 2023 - Mar 2024

- Research on spatio-temporal and network methods for mobility and infrastructure data; methodology published in a peer-reviewed journal.

AlmaViva S.p.A. | Trento, Italy

Data Scientist | Feb 2023 - Feb 2024

- Built ML and analytical pipelines on sensor and behavioural data, from acquisition to deployment (Python, Scikit-learn).

University of Trento; University of Toronto; Nanyang Technological University | IT / CA / SG

Research in Experimental Design and Signal Processing | Sep 2015 - Mar 2020

- Designed and ran controlled experiments and processed high-dimensional time-series data (EEG/ERP, eye-tracking, pupillometry) in Python and R, resulting in peer-reviewed publications and international conference presentations; two international research scholarships.

EDUCATION

Master of Science, Data Science | University of Trento, Italy | Sep 2020 - Dec 2023

Final grade: 110/110 with honours

Master of Science, Psychology (Neuroscience) | University of Trento, Italy | Sep 2016 - Mar 2019

Final grade: 110/110 with honours. Strong focus on statistics and experimental methodology.

Bachelor of Science, Cognitive Psychology | University of Trento, Italy | Sep 2013 - Sep 2016

Final grade: 108/110

PUBLICATIONS

- Garbin S., Venturoso L., Todescato M. (2026). Automatic weld seam segmentation for industrial quality control: a comparison of RGB and polarimetric imaging with CNN and transformer architectures. Under review.
- Venturoso L., Garbin S., Steiner D., Matt D. T. (2025). AI smart workstation for industrial quality control. *Industry 4.0 Science*, 41(5), 128-134.
- Venturoso L., Usmani M., Nanni R., Napolitano M. (2025). LTS-BikePlan: a data-driven tool for enhancing cycling infrastructure and safety. *Journal of Urban Technology*.
- Venturoso L., Gabrieli G., Truzzi A., Azhari A., Setoh P., Bornstein M. H., Esposito G. (2019). Effects of baby schema and mere exposure on explicit and implicit face processing. *Frontiers in Psychology*, 10, 2649.

AWARDS AND SELECTED CONFERENCES

- Winner, Premio Tesi Citta di Bolzano 2024; Finalist, Alpine Convention Young Academics Award 2024.
- Presentation, AI Smart Workstation for Industrial Quality Control, WGAB-Forschungsseminar 2025, TU Hamburg, Germany.
- Presentation, International Attachment Conference 2019, Vancouver, Canada; Poster, Nanyang Technological University, Singapore, 2016.